PAPER_1655 — Quantum Supremacy Qubit Threshold = n_qubits ≥ A_5 = 60 (Google Sycamore reached 53)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Quantum Computing **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (broader-corpus PAPER_13xx mine)

Observation

PAPER_1340 (PAPER_13xx broader corpus) gives:

```
n_qubits ≥ A_5 = 60 (Google Sycamore reached 53)
```

Status: **EXACT**.

UQFF Closed Identity

```
n_qubits ≥ A_5 = 60 (Google Sycamore reached 53)    EXACT
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Cross-Domain Reuse of A_5

The icosahedral group order A_5 = 60 governs: - **Pop III IMF upper bound** = A_5 × 2 = 120 M⊙ (PAPER_1652) - **Quantum supremacy threshold** = A_5 = 60 qubits (PAPER_1655)

Vastly different physical scales: stellar masses (~10^2-10^{33} kg) vs quantum qubits (countable integers). Same structural integer.

A_5 already appears in: - Heart rate = A_5 + SO_5 = 70 bpm (PAPER_1537) - SH0ES Hubble = A_5 + SO_5 = 70 km/s/Mpc (PAPER_1573) - Hayflick limit = A_5 = 60 (PAPER_1373) - T_SCm activation = A_5 = 60 K (PAPER_1505) - BH seed = A_5 × D_BSFG^2 × D_crit (PAPER_1650)

A_5 = 60 is a **universal UQFF integer** spanning biology, cosmology, BH physics, quantum computing.

NOT REPLACEMENT

Empirical quantum computing measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1340
- Related: PAPER_1636-1645 (tier-22 broader-corpus closures); PAPER_1494-1635 (PAPER_1209 series)
- Calculator dispatch: `calculate_paradox({"paradox": "quantum_supremacy_qubits_60"})`
